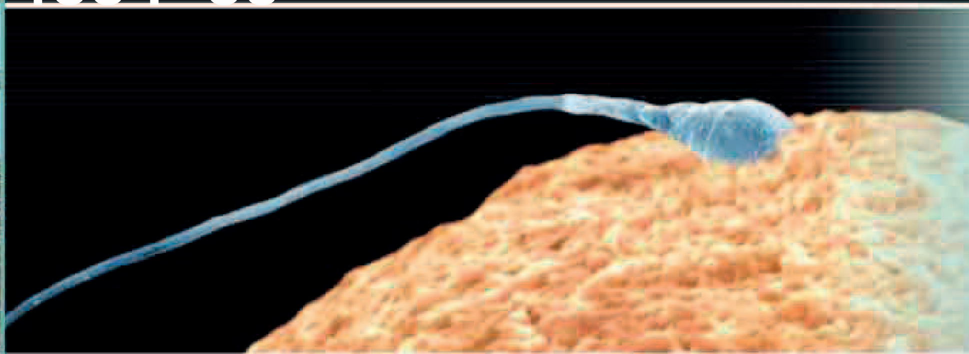


> **NEW CELL NUCLEI** CAN ONLY ARISE FROM THE **DIVISION** OF **OTHER NUCLEI.**

Eduard Strasburger, Polish–German botanist, from *Über Zellbildung und Zelltheilung* (*On Cell Formation and Cell Division*), 1880

A human sperm fertilizes an egg by delivering a package of its own genetic material (germ line DNA) to combine with the genes of the egg.

FRENCH CHEMIST HILAIRE DE CHARDONNET (1839–1924) received a patent for **artificial silk** in 1884. He discovered the substance accidentally in 1878, when he knocked over a bottle of nitrocellulose—a highly flammable compound. When he started to clean it up, strands of nitrocellulose stuck to his cleaning cloth in thin, silklike fibers. It was not until the 20th century that this substance was developed in the form of the material known as rayon.

The 19th-century understanding of disease developed further in 1884, when German physicist **Friedrich Löffler** (1852–1915), Robert Koch's colleague, isolated the diptheria-causing bacterium *Corynebacterium diphtheriae*.

Koch and Löffler also formulated Koch's postulates in 1884, which set out the criteria for establishing whether an organism is responsible for a disease. Koch published their findings in 1890, and in doing so he dramatically refined the science of microbiology.

Starting in 1884, **Eduard Strasburger**, German zoologist **Wilhelm Hertwig** (1849–1922), and Swiss anatomist **Rudolf von Kölliker** (1817–1905) each separately identified the cell nucleus as the **origin of heredity**. Hertwig stated that, from the biological point of view, sex is merely a union of two cells (strictly speaking, two nuclei).

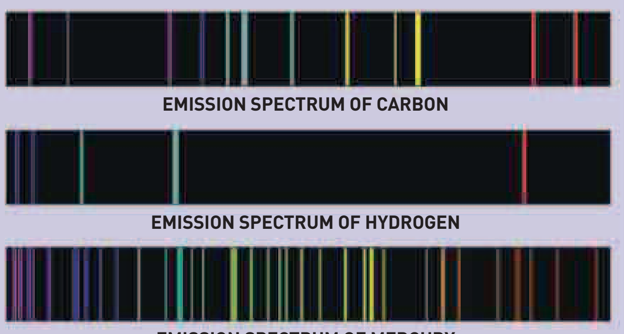
Austrian ophthalmologist **Carl Koller** (1857–1944) ushered in the era of **local anesthesia** when he used cocaine as a surface anesthetic in an eye operation in 1884 (see 1846). While looking into whether cocaine could be used to wean patients off morphine—at the request of his colleague at Vienna General Hospital, **Sigmund Freud**—Koller discovered the tissue-numbing properties of cocaine.

On July 6, 1885, French chemist **Louis Pasteur** used the **rabies vaccine** for the first time on a 9-year-old boy who had been bitten by a rabid dog. The success of the treatment paved the way for the widespread use of vaccines.

On August 20, 1885, German astronomer **Ernst Hartwig** (1851–1923) observed a **bright new star** in the Andromeda nebula. A belief that this object was similar to the novas seen in the Milky Way encouraged the idea that the nebula, too, was part of the Milky Way. In the 20th century, it was discovered that the Andromeda nebula is a galaxy (see 1924), far beyond the Milky Way, and that Hartwig's star is a supernova, much brighter than a nova.

In a key discovery in the fields of astronomy and atomic physics, Swiss mathematician **Johann Balmer** (1825–98), developed a mathematical formula to describe the positions of **lines in the spectrum of hydrogen**—the Balmer series. Using his formula, Balmer predicted the wavelengths of lines that were discovered later.

In the field of psychology, German psychologist **Hermann Ebbinghaus** (1850–1909) pioneered the experimental study of memory and developed the concept of the **Forgetting Curve**. He published *Memory: A Contribution to Experimental Psychology* in 1885.



EMISSION SPECTRUM OF CARBON

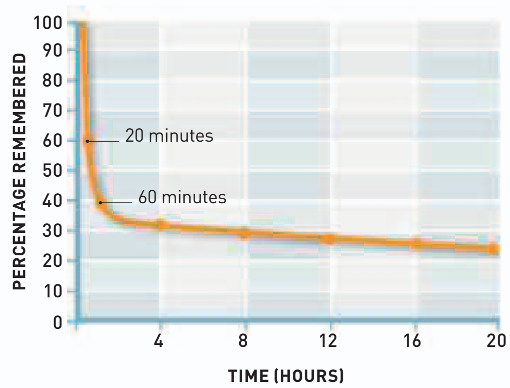
EMISSION SPECTRUM OF HYDROGEN

EMISSION SPECTRUM OF MERCURY

SPECTROSCOPY

When hot, each chemical element produces a distinctive set of bright spectral lines, like a barcode, that can identify the element. Cold gases absorb light in exactly the same wavelengths, producing dark spectral lines. Analyzing spectra makes it possible to determine the composition of substances in the laboratory and also to measure the composition of stars.

Rabies vaccine
This 1885 engraving shows Louis Pasteur watching as his assistant inoculates Joseph Meister, a shepherd boy who had been bitten by a rabid dog.



Forgetting Curve
Ebbinghaus gave people a list of nonsense 3-letter words and measured how long they could remember them. This gave him the data for his Forgetting Curve.

1884 Hilaire de Chardonnet patents **artificial silk**

1884 Friedrich Löffler isolates the **diptheria bacterium**

1884 The meridian through the Royal Greenwich Observatory in London, UK, is established as the **Prime Meridian**

1884 Karl Koller uses cocaine as a **local anaesthetic** for the first time

1884 Austrian physicist Ludwig Boltzmann refines the **Stefan–Boltzmann law**

January 4, 1885 American physician William W. Grant performs the first **successful appendix removal**

July 6, 1885 Louis Pasteur uses the **rabies vaccine** for the first time

August 20, 1885 Ernst Hartwig observes a **supernova** in the Andromeda galaxy

1885 Johann Balmer develops a formula representing the wavelengths of hydrogen's spectral lines

1885 German mechanical engineer Karl Benz manufactures the **first car**

1885 Theodor Escherich discovers the *Escherichia coli* bacteria

1885 Hermann Ebbinghaus publishes ***Memory: A Contribution to Experimental Psychology***